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TECHNO-METER

TURNING WASTE INTO WISDOM FOR
PHYSICS LABS

Group number : 30207 **New Cairo STEM School**

INTRODUCTION

Problem Statement :

- Inefficient E-waste recycling in Egypt
- Heavy metal pollution in air and Nile water
- Lack of affordable motion-measurement tools in school labs
- Discarded devices labeled as “waste” despite high value

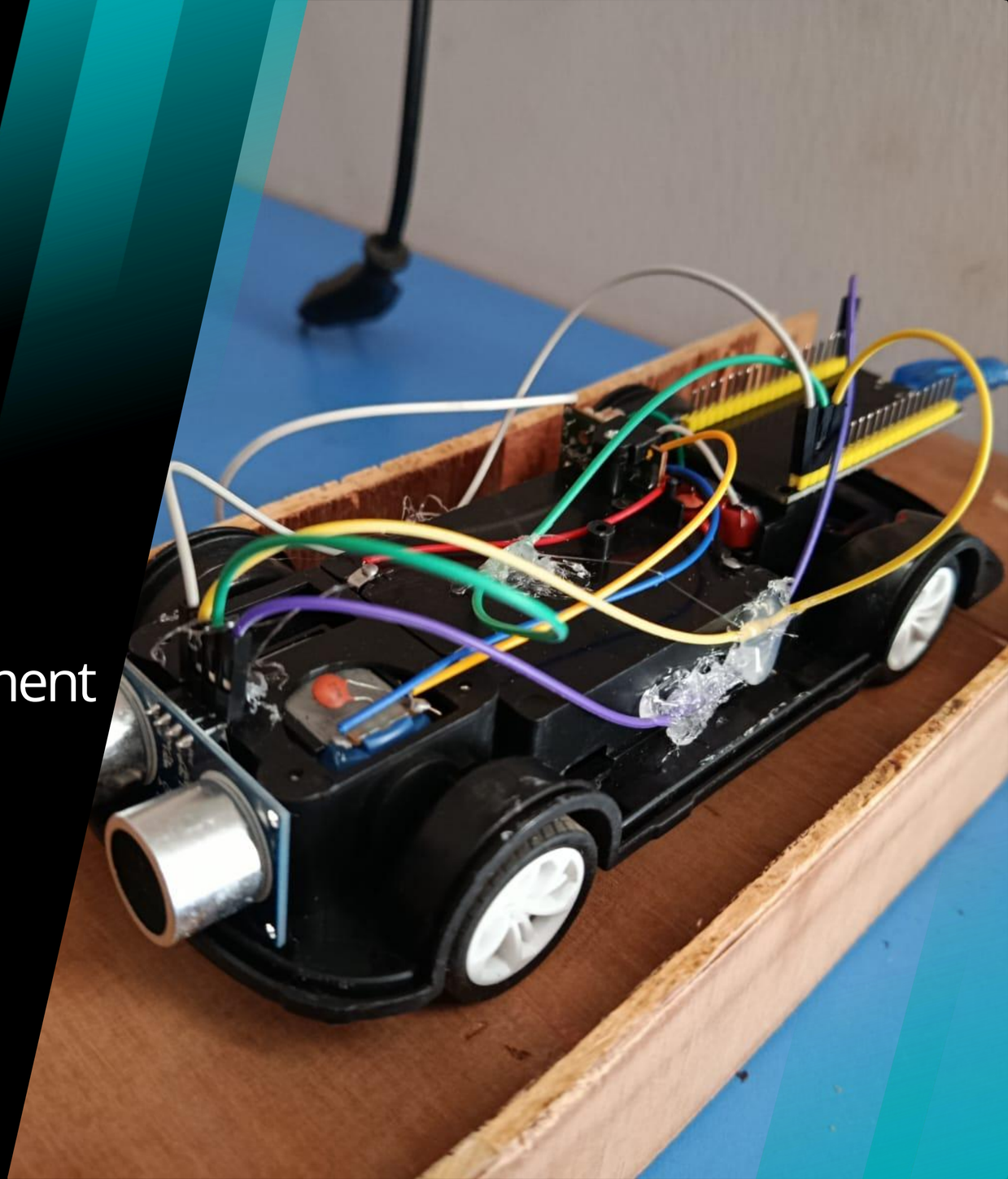
Research Background :

- PCB recycling: effective but complex and costly
- Ultrasonic sensors: accurate distance measurement
- Previous solutions lack educational accessibility



DESIGN REQUIREMENTS & CONSTRAINTS

- Built mainly from discarded devices
- Multi-functional or multi-lab usage
- No hazardous or radioactive materials
- Safe for students
- Error percentage ≤ ±5%
- Improve scientific & technological environment

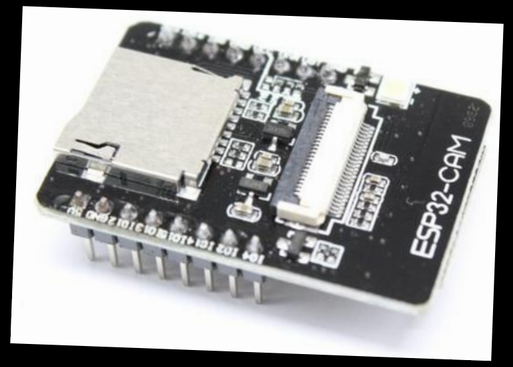


OUR MATERIALS



01

OLD TOY CAR



03

ESP-32 CAM



05

JUMPER WIRES



02

ULTRA-SONIC SENSOR

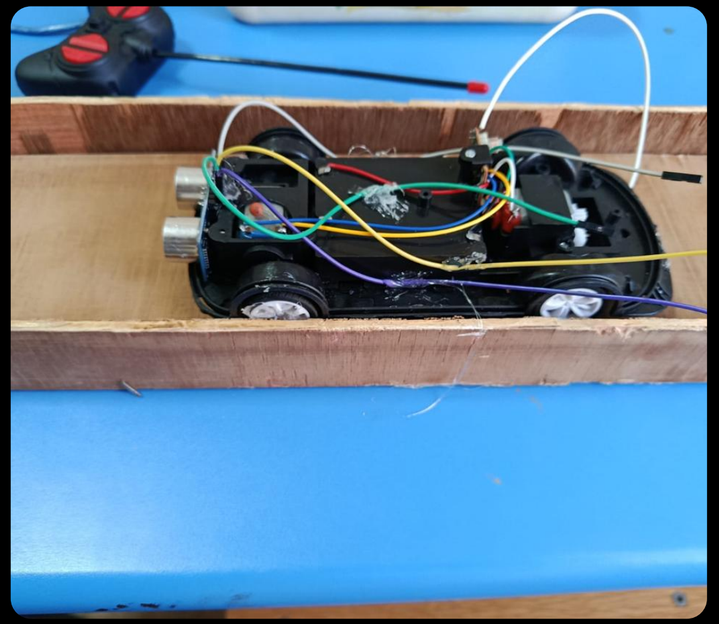


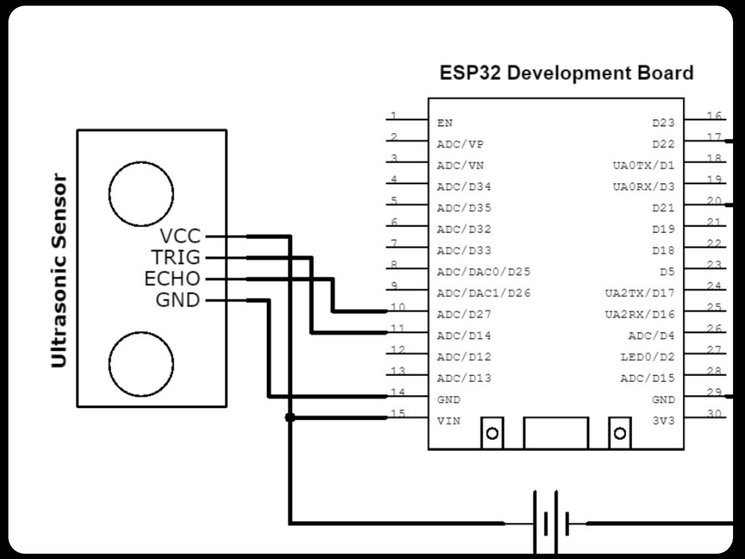
04

DISCARDED WOOD

METHODOLOGY (CONSTRUCTION STEPS)

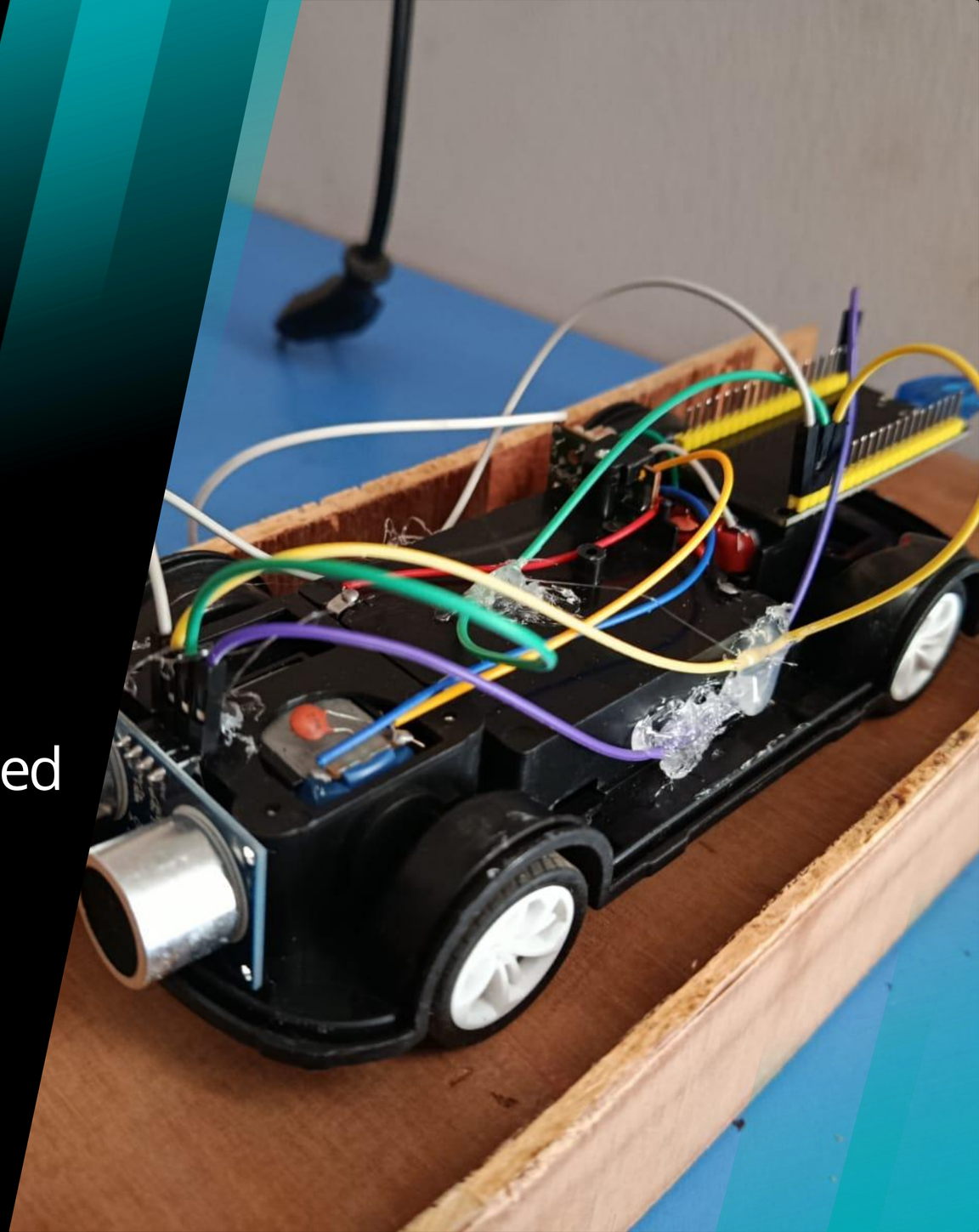
- EXTRACT MICROCONTROLLER FROM DISCARDED CAMERA
- CALIBRATE ULTRASONIC SENSOR
- CONNECT COMPONENTS USING OLD WIRES
- PROGRAM ESP32-CAM
- DEVELOP IOT WEBSITE (HTML, CSS, JAVA)
- ASSEMBLE SYSTEM ON TOY CAR





TEST PLAN & SAFETY

- Safety precautions:
  - Gloves
  - Lab coats
- Calibration under lab conditions
- Manual vs sensor measurements
- Controlled environment
- Taking the results from the website to be analyzed



OUR RESULTS

NEGATIVE RESULTS

- Initial experiments used a Hall effect sensor to measure distance
- The sensor showed inconsistent readings and skipped pulses
- Average measured distance was ≈ 0.70 m instead of 1.00 m
- Error percentage reached ~30%, exceeding acceptable limits
- Causes of error:
- Incompatibility with ESP32-CAM
- Magnet size affected pulse detection
- Mechanical instability of the rotating wheel

OPTIMIZED RESULTS & SYSTEM PERFORMANCE

- Hall effect sensor replaced with an ultrasonic sensor
- Distance measured using Time-of-Flight (TOF) principle
- Manual distance average: 1.00 ± 0.02 m
- Sensor distance average: 0.98 ± 0.02 m
- Final error percentage reduced to ≈ ±2%
- Speed and acceleration calculated accurately
- Stable graphs suitable for physics lab experiments

Table 1: Distance readings

Distance	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Averag e
Manual measuring value	1.02 m	1 m	1.01 m	0.98 m	0.99 m	1m
Measured value by sensor	1 m	0.98 m	0.99 m	0.96 m	0.97	0.98 m

Table 2 :Test Plan Readings

Quantity	Trial 1	Trial 2	Trail 3	Trial 4	Trial 5
Distance	1 m	0.98 m	0.99 m	0.96 m	0.97 m
Time	1.72 s	1.67 s	1.75 s	1.6 s	1.5 s
Avg. Speed	0.581 m/s	0.587 m/s	0.566 m/s	0.600 m/s	0.647 m/s
Avg . Acceleration	1.5	1.48	1.56	1.6	1.58

ANALYSIS

- Initial measurements using the Hall effect sensor showed large deviations
- The measured distance (≈ 0.70 m) differed significantly from the actual distance (1.00 m)
- Error percentage reached ~30%, which is unacceptable for physics lab applications
- Scientific interpretation:
- Hall effect sensors depend on accurate magnetic pulse detection
- Pulse skipping occurred due to:
- Magnet size mismatch
- Mechanical instability
- Sensor–microcontroller incompatibility
- The Hall effect sensor was replaced by an ultrasonic sensor

SYSTEM ARCHITECTURE

- Ultrasonic sensor measures distance (TOF method)
- ESP32-CAM processes data
- Website displays:
  Tables
  Graphs
- IoT-based real-time monitoring

WORKING PRINCIPLE

- Ultrasonic wave emitted → reflected → received
- Time of flight used to calculate distance
- Speed and acceleration derived mathematically
- Environmental factors considered (temperature

LEARNING TRANSFER

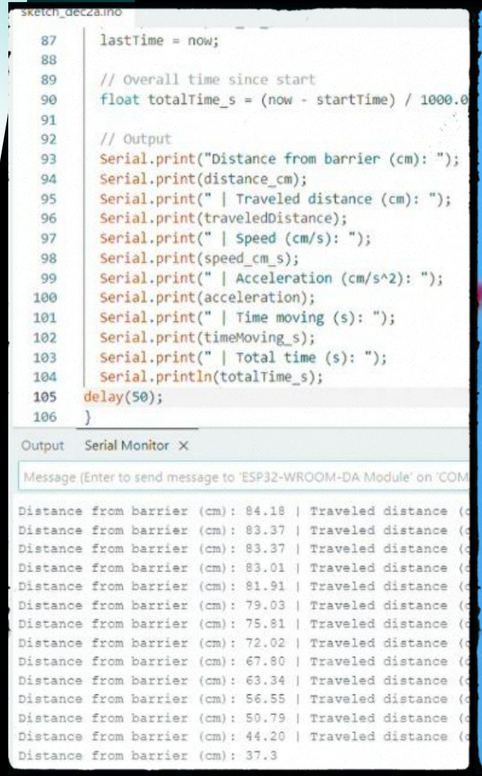
Subject	Learning Transfer
PH.2.03	Clarified the laws of electrodynamics and helped us determine the suitable voltage for each component using Ohm's law (V=
PH.2.06	Facilitated the usage of the Hall effect sensor, as it discussed magnetic fields and helped in determining the suitable distar between the sensor and the magnet.
English	Help in conducting this research as a whole, in writing the poster and portfolio academically.
MA.2.01	Facilitated the graphing of the equations used and ensured the correctness of the results.

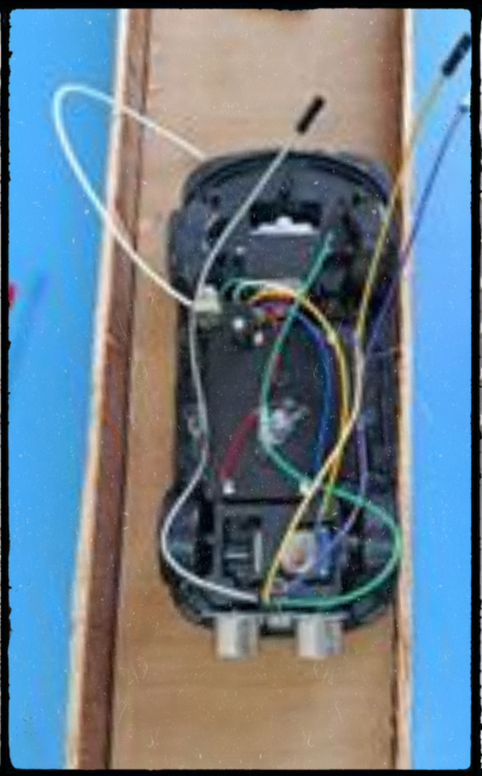
CONCLUSION

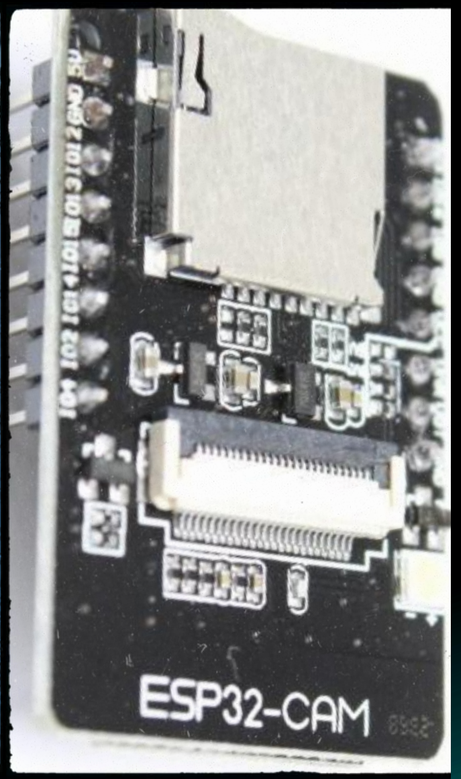
This project demonstrates that recycled electronic waste can be transformed into a reliable and educational scientific tool. Through iterative testing and analysis, initial limitations—such as the failure of the Hall effect sensor—were identified, scientifically interpreted, and successfully addressed by adopting ultrasonic sensing technology. This optimization reduced the measurement error from approximately 30% to ±2%, meeting and exceeding the project's design requirements.

The final prototype accurately measures distance, speed, and acceleration, and provides real-time data visualization through an IoT-based website, making it highly suitable for physics laboratory applications. Beyond technical performance, the project highlights the potential of sustainable engineering to enhance the scientific and technological environment while simultaneously reducing electronic waste.

Overall, the Techno-meter confirms that low-cost, recycled components, when combined with sound scientific principles and systematic analysis, can deliver accurate, reliable, and environmentally responsible solutions for educational use.







RECOMMEDATION

There is no perfect prototype, and our prototype is not exception. We recommend some features and attributes that can make major effects in the project that could help in improving the application and efficiency of our project. To turn this project to a final product several suggestions should be considered as follows:

I. Improve Measurement Accuracy: using better, more accurate sensors along with the processing unit is critical for a larger scale building, considering its scarcity regarding small scale prototypes.

II. Reinforce Mechanical Components: Strengthen the car chassis and overall car body and the track, ensure stable performance and reduce errors considering high-speed testing

III. Adopt adaptive design: use flexible sensor and hardware modules for replacements and customizations for educational lab experiments.

IV. Increase Sustainability Impact: adopt a well-studied technique of recycling waste materials (using a well-studied plan of which materials will be recycled and how to ensure having the best impact on earth)

V. Ensure Reliability and Safety: to make a final product out of our prototype, it must meet several safety requirements and standards to withstand any crisis and save students from any fatal error.



LITERATURE CITED

1.ABDEL-SATAR, A. M., ALI, M. H., & GOHER, M. E. (2017). INDICES OF WATER QUALITY AND METAL POLLUTION OF NILE RIVER, EGYPT. EGYPTIAN JOURNAL OF AQUATIC RESEARCH, 43(1), 21-29.

2.Aliew, F. (2022). An approach for precise distance measuring using ultrasonic sensors. Engineering Proceedings, 24(1), 8.

3.Cameron, N. (2023). ESP32 microcontroller. In ESP32 Formats and Communication: Application of Communication Protocols with ESP32 Microcontroller (pp. 1-54). Berkeley, CA: Apress.

4.Ghanem, S., Ferrini, S., & Di Maria, C. (2023). Air pollution and willingness to pay for health risk reductions in Egypt: A contingent valuation survey of Greater Cairo and Alexandria households. World Development, 172, 106373.

5.Jeon, S., & Seo, D. (2019). Smart parking system based on an ultrasonic sensor and bluetooth low energy in the internet of things. J. Syst. Manag. Sci, 9, 91-110.

6.Salikhov, R. B., Abdrakhmanov, V. K., & Safargalin, I. N. (2021, November). Internet of things (IoT) security alarms on ESP32-CAM. In Journal of Physics: Conference Series (Vol. 2096, No. 1, p. 012109). IOP Publishing.

7.Tan, Q., Liu, L., Yu, M., & Li, J. (2020). An innovative method of recycling metals in printed circuit board (PCB) using solutions from PCB production. Journal of Hazardous Materials, 390, 121892.

THANK YOU

http://feronabil2008-del.github.io/website-techno-meter/.COM

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